

# Sensitivity analysis

Project: Reproducible AI Research Study

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Ablation studies

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Statistical tests and assumptions

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Sampling variability

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace the worked case for Sensitivity analysis. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Sensitivity analysis. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply a new case independently

Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

### My reasoning and evidence

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## 6. W2 Trace the worked case

Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

### My reasoning and evidence

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## 7. W3 Audit the limit

A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

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## 8. W4 Apply a new case independently

Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

**My reasoning and evidence**

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## 9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

### My reasoning and evidence

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## 10. Project checkpoint and reflection

Add a labelled sensitivity analysis evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of sensitivity analysis. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### My project change, supporting evidence and remaining limit

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## 11. Feedback, return check and learning record

## Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

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**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Sensitivity analysis

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

Scores are 0.4,0.6,0.8. Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

## Score decisions under inclusive cutoffs

|            | Score 0.4 | Score 0.6 | Score 0.8 |
|------------|-----------|-----------|-----------|
| Cutoff 0.5 | No        | Yes       | Yes       |
| Cutoff 0.7 | No        | No        | Yes       |
| Cutoff 0.6 | No        | Yes       | Yes       |

Read a cell using both its row and column labels.

## Sensitivity analysis

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Hold scores fixed

Keep the same three supplied scores.



#### 2. Apply cutoff 0.5

0.6 and 0.8 meet the boundary.



#### 3. Apply cutoff 0.7

Only 0.8 meets the new boundary.

### CHANGE ONE THING

Set the cutoff to exactly 0.6.

Explain the changed result using the matching lesson.

Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.